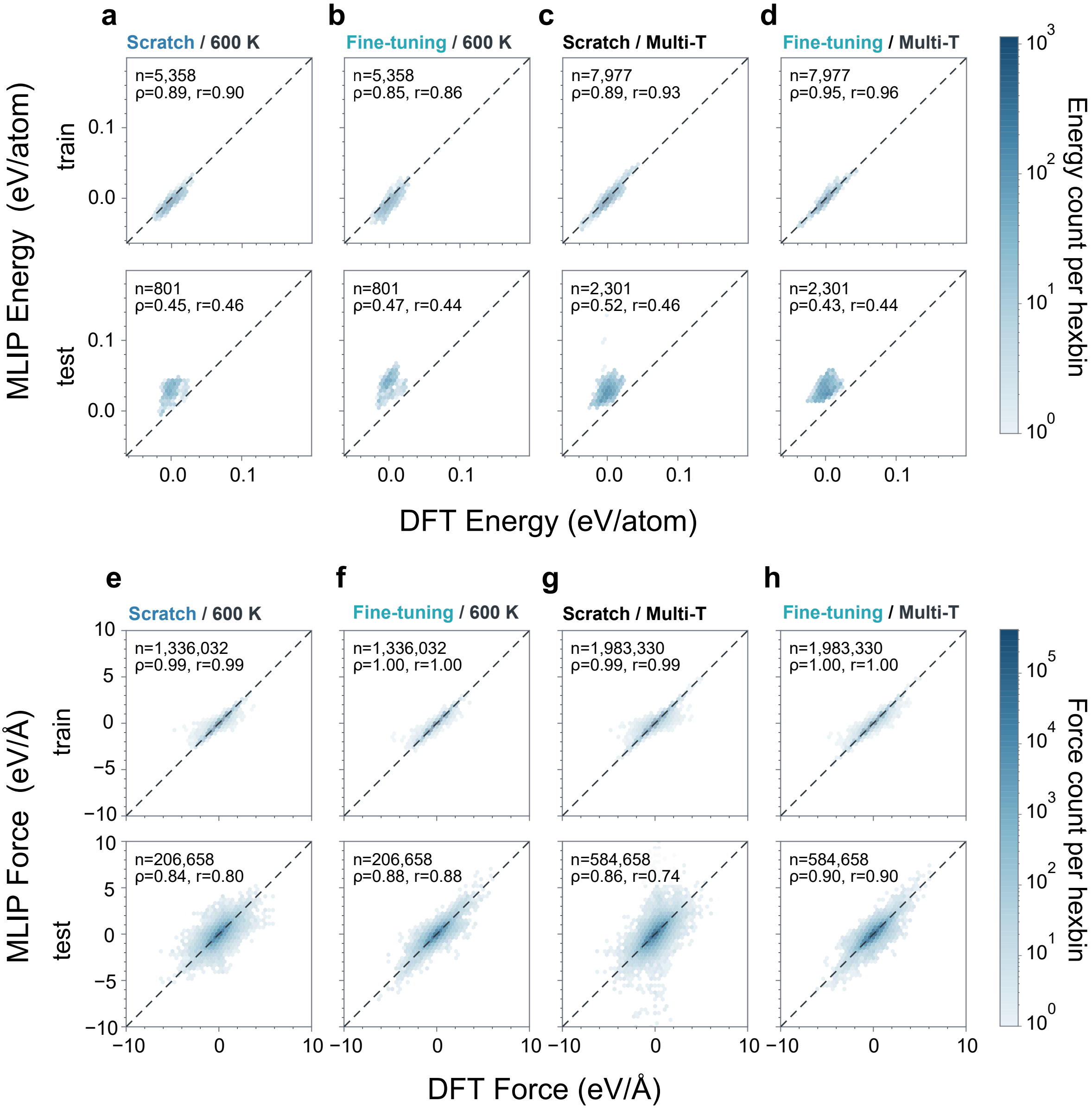

a
b
c
d
Scratch / 600 K
Fine-tuning / 600 K
Scratch / Multi-T
Fine-tuning / Multi-T
n=5,358
ρ=0.89, r=0.90
n=5,358
ρ=0.85, r=0.86
n=7,977
ρ=0.89, r=0.93
n=7,977
ρ=0.95, r=0.96
n=801
ρ=0.45, r=0.46
n=801
ρ=0.47, r=0.44
n=2,301
ρ=0.52, r=0.46
n=2,301
ρ=0.43, r=0.44
train
test
MLIP Energy (eV/atom)
DFT Energy (eV/atom)
Energy count per hexbin
e
f
g
h
n=1,336,032
ρ=0.99, r=0.99
n=1,336,032
ρ=1.00, r=1.00
n=1,983,330
ρ=0.99, r=0.99
n=1,983,330
ρ=1.00, r=1.00
n=206,658
ρ=0.84, r=0.80
n=206,658
ρ=0.88, r=0.88
n=584,658
ρ=0.86, r=0.74
n=584,658
ρ=0.90, r=0.90
MLIP Force (eV/Å)
DFT Force (eV/Å)
Force count per hexbin